Stage 13 讲稿（中英对照）

一、为什么要做 Stage 13

前面的实验留下了一个矛盾。读出工具在中间层看不到作者名字，这是 Stage 10 的结果；但替换实验说明，到 L48 信息已经在了，这是 Stage 11 和 12 的结果。那中间层里到底是什么？这个问题用原来的工具回答不了，因为原来的工具只能读“词”，而问题恰恰在于信息可能不是以词的形式存在的。所以 Stage 13 必须换测法：读年份、换问法、换标题。

Our earlier experiments left a contradiction. The readout tools cannot see author names in the middle layers; that is Stage 10. Yet patching shows the information is already there by L48; that is Stages 11 and 12. So what is in the middle layers? The original tools cannot answer this, because they can only read "words," and the whole question is whether the information exists in some other form. So Stage 13 has to measure differently: read out the year, change the question, swap the title.

二、控制范围

偏离主问题的风险是真的。Stage 13 已经有 M1–M4、X1–X7 十几个实验，有些离主问题比较远，比如“能不能从标题读出引用数”，本身跟“模型知不知道作者”关系不大。所以我们定了一条规则：每个实验都必须能回答一句“这对主问题有什么用”，答不上来的就降级成附录或者先停。

The risk of drifting is real. Stage 13 already has more than ten experiments, M1 to M4 and X1 to X7, and some are fairly far from the main question; whether citations can be read from the title, for example, says little by itself about whether the model knows the authors. So we set one rule: every experiment must answer "what does this do for the main question?" Anything that cannot is moved to an appendix or paused.

三、M1：正确名字排不上，那排第一的是谁？

M1 接的是 Stage 10。提示、名字位置、读出工具（Jacobian lens 和 logit lens），以及真标题对假标题的对比，全部照搬，只换了一处：Stage 10 看的是正确名字排第几，M1 看的是排第一的是什么词。Stage 10 只告诉我们正确名字排得很低，没告诉我们它前面挤着什么，M1 就把那个位置上的东西翻出来看；再用 Stage 10 的假标题设计，看排第一的词会不会跟着标题变，判断它跟论文有没有关系。

M1 builds on Stage 10. The prompt, the name position, the readout tools (Jacobian lens and logit lens), and the true-versus-false-title contrast are all reused. Only one thing changes: Stage 10 asks where the correct name ranks, while M1 asks what ranks first. Stage 10 told us the correct name ranks low, but not what is crowding in front of it; M1 opens that up. Using Stage 10's false-title design, we also check whether the top token changes with the title, which tells us whether it has anything to do with the paper.

结果是：在 L24 到 L48，Jacobian lens 排第一的全是句号、换行这类符号；L51、L52 起出现 Robert、Zhang 这样的常见名字。正确名字在 L52 只占 0.1%，L62 也只占 2%。logit lens 在中间层几乎全是无意义的 token，到 L62 排第一的才有 87% 是像名字的 token，但不一定是正确名字。所以读出工具在中间层看到的主要是格式，这就是 Stage 10 在那里看不到名字的原因。

The result: from L24 to L48, the Jacobian lens ranks sentence-end punctuation and line breaks first in every case; common names like Robert and Zhang appear from L51 and L52. The correct name is top only 0.1% of the time at L52 and 2% at L62 for this lens. The logit lens shows mostly meaningless tokens in the middle layers, and only at L62 are 87% of its top tokens name-like, though not necessarily the correct name. So what the readout tools see in the middle layers is mainly format, which is why Stage 10 cannot see names there.

四、M2：读不出作者，那读得出论文的其他信息吗？

M2 接的是 Stage 11 的 probe。位置（标题最后一个 token）、方法（简单线性模型）、按作者团队切分的五折，以及 Stage 1 的 target / control 标签，全部沿用。只换了一处：Stage 11 的 probe 读作者，信号很弱，只有 +0.05 到 +0.08；M2 改读年份、引用数、领域、认不认得出。逻辑是：如果标题位置存的是“这是哪篇论文”，那除了作者，这篇论文的其他属性也应该读得出来。选标题最后一个 token，是因为提示里年份写在标题后面，这个位置还没看到年份，不会泄题。“认不认得出”直接用 Stage 1 的分组当答案，引用数也跟 Stage 10 发现的“引用越高、信号越强”接上。

M2 builds on the Stage 11 probe. The position (the last title token), the method (a simple linear model), the five folds split by author team, and the Stage 1 target/control labels are all reused. Only one thing changes: the Stage 11 probe reads authors and the signal is weak, +0.05 to +0.08; M2 reads year, citations, field, and recognised-or-not instead. The logic: if the title position holds "which paper this is," other properties of the paper should be readable too. We use the last title token because the year appears after the title in the prompt, so this position has not seen it yet and cannot leak it. Recognised-or-not uses the Stage 1 groups as the answer, and citations connect to Stage 10's finding that more-cited papers carry a stronger signal.

probe 是怎么工作的

先说输入。模型在第 24 层，标题最后一个 token 上的内部状态是一串数字，比如 [0.3, −1.2, 0.8, 0.05, ……]，一共 5,120 个。每篇论文取一串，9,108 篇就是 9,108 串。我们看不懂这些数字各代表什么，所以需要一个工具来翻译。这个工具就是一个打分公式：给 5,120 个数各配一个权重，乘起来再加总，得出答案，也就是“猜的年份 = 第 1 个数 × 权重 1 + …… + 第 5,120 个数 × 权重 5,120 + 常数”。权重一开始不知道，靠训练学出来：拿几千篇知道真实年份的论文，不断调整权重，让猜出来的年份尽量接近真实年份。学好之后，拿没见过的论文去测，只给它那串数字，看它猜得准不准。

First, the input. At layer 24, the model's internal state at the last title token is a list of numbers, like [0.3, −1.2, 0.8, 0.05, …], 5,120 in total. Each paper gives one list, so 9,108 papers give 9,108 lists. We cannot tell what each number means, so we need a tool to translate. The tool is a scoring formula: give each of the 5,120 numbers a weight, multiply and add them up. In other words, predicted year = number 1 × weight 1 + … + number 5,120 × weight 5,120 + a constant. The weights are learned: we take thousands of papers whose year we know and keep adjusting the weights until the predictions are as close as possible. Then we test on papers it has never seen, giving it only the list of numbers, and check how accurate it is.

每一层单独做一次：取第 16 层的那串数字，训练一个工具，测准确度；再取第 24 层的，重新训练一个新工具，再测。每层一个独立的工具，最后把每层的准确度连起来就是一条曲线。曲线告诉我们，模型在哪一层开始“知道”这篇论文的年份，以及越往后知道得越清楚。

We do this separately for each layer: take layer 16's numbers, train a tool, measure accuracy; take layer 24's, train a fresh tool, measure again. One independent tool per layer. Joining the accuracies

gives a curve, which tells us at which layer the model starts to "know" the paper's year, and how much clearer it gets in later layers.

对照与结果

为了公平，我们加了几个对照。第一，把答案打乱再训练，看工具会不会瞎猜也能猜对，结果不会：领域准确率从 0.69–0.72 掉到 0.13。第二，只给标题文字，不看模型内部，用 BERT 去猜，看光靠文字能猜出多少。第三，在冒号位置做同样的测试作为阳性对照，因为那里已经看到年份，年份 R^2 达到 0.98，说明方法能测出真实存在的信息。

To keep it fair, we added controls. First, we shuffle the answers and retrain to see whether the tool can succeed by chance; it cannot: field accuracy drops from 0.69–0.72 to 0.13. Second, we give BERT only the title text, without looking inside Qwen, to see how much the text alone predicts. Third, as a positive control, we run the same test at the colon position, which has already seen the year; year R^2 reaches 0.98 there, showing the method detects information that is truly present.

结果：从标题最后一个 token 就能读出年份（R^2 从 0.39 升到 0.56）、引用数（0.28 升到 0.53）和领域，而且从 L16 就开始了，比名字早很多。光看标题文字的 BERT，年份只有 0.24、引用数只有 0.19，明显更弱。但有一项例外："认不认得出"，Qwen 是 0.67，BERT 是 0.63，基本追平，所以这一项光看标题文字就能猜个大概，我们不拿它当"认出论文"的证据。另外要说明，Qwen 比 BERT 强，也可能只是因为 Qwen 更会理解文字，所以这只能作参考，不能单独证明"认出了论文"。

The result: from the last title token we can read the year (R^2 rising from 0.39 to 0.56), citations (0.28 to 0.53), and field, starting from L16, well before names. BERT on title text alone gets only 0.24 for year and 0.19 for citations, clearly weaker. One exception is recognised-or-not: Qwen 0.67, BERT 0.63, nearly equal. So the title text alone predicts it fairly well, and we do not use it as evidence of recognition. Also, Qwen beating BERT may simply mean Qwen understands text better, so this is supporting evidence only and cannot by itself prove the model recognised the paper.

五、M3 和 X1：标题不变，换问法

M3 沿用 Stage 10–11 的核心设计：只换一个东西，其他全部不变，但这次反过来换。Stage 10–11 是问题不变、换标题，看作者分数变不变；M3 是标题不变、换问题，问作者、问年份、问领域，看标题位置的状态变不变。前面证明了"换标题，信息就变"，说明信息跟论文绑定；M3 补上另一半："换问题，信息变不变"。如果不变，说明中间层存的是论文本身，跟任务无关；如果变，说明存的是"要怎么回答"。

M3 reuses the core design of Stages 10 and 11: change one thing and hold everything else fixed, but this time in reverse. Stages 10 and 11 held the question fixed and swapped the title, checking whether author scores changed. M3 holds the title fixed and swaps the question, asking about authors, year, or field, and checks whether the state at the title position changes. Earlier we showed that swapping the title changes the information, so it is tied to the paper. M3 supplies the other half: does changing the question change it? If not, the middle layers hold the paper itself, independent of the task; if so, they hold "how to answer."

X1 更进一步，直接用 Stage 11 的替换实验：把"问年份"那遍的标题状态，替换进"问作者"那遍，用 Stage 11 同一个恢复率公式，看作者分数能不能恢复。能恢复，就说明"问年份时读到的论文信息"和"问作者时用的"是同一份。

X1 goes further and reuses the Stage 11 patching directly: we take the title state from the year-question run, patch it into the author-question run, and use the same recovery formula as Stage 11 to see

whether author scores come back. If they do, the paper information read under the year question is the same information used under the author question.

结果：在 L24 到 L48，同一篇论文换问法，标题位置状态的余弦相似度在 0.95 以上，移动的距离只有换一篇论文的 15% 到 27%。但要说清楚一个关键限制：按预注册补充 2，M3 要比 L4 高才算数，因为 L4 几乎只编码了标题文字。结果年份的迁移比 T − T(L4) 是 −0.03 到 −0.06，引用数也是负的，领域和 L4 分不开；身份指数 I − I(L4) 在所有层都小于 0。也就是说，中间层并不比 L4 更跨问法一致，余弦高、移动小，很大程度上是因为三种问法里的标题文字完全一样，不能单独证明论文身份。X1 的结果：把问年份、问领域时的标题状态换进问作者那遍，L24 到 L48 恢复作者分数和用问作者的差别都在预注册的 ±0.1 以内；减掉随机论文 C 的结果后判定不变；L48 的作者减年份是 +0.019，很小但区间不含 0，说明那一层有一点真实的任务依赖。冒号位置正好相反：换问法时它的变化是换论文的 3 到 5 倍，存的是任务。

The result: from L24 to L48, the same paper under different questions keeps a cosine similarity above 0.95 at the title position, moving only 15 to 27 percent as far as switching papers. But one key limit must be stated: under pre-registered addendum 2, M3 only counts if it beats L4, which encodes little beyond the title text. For year, T − T(L4) is −0.03 to −0.06; citations are also negative and field is indistinguishable from L4; the identity index I − I(L4) is below 0 at every layer. So the middle layers are no more consistent across questions than L4; the high cosine and small shift largely reflect identical title text in the three prompts and cannot by themselves show paper identity. X1: patching in the title state from the year or field question restores author scores within the pre-registered ±0.1 of the author question at L24–48; subtracting the random paper C changes no verdict; at L48 author minus year is +0.019, small but with an interval excluding 0, a little real task dependence there. The colon position is the opposite: it moves three to five times more when the question changes than when the paper changes, so it holds the task.

六、三个假设：预测与结果

我们一开始列了三个假设，每个都对 M1、M2、M3 给出不同的预测。H1 说中间层存的是任务和格式：“我要输出一个作者列表”。H2 说存的是论文身份：“这是哪篇论文”。H3 说作者其实已经在了，只是不在词表方向上，所以读出工具看不到。

We started with three hypotheses, each making different predictions for M1, M2 and M3. H1 says the middle layers hold the task and format: "I am about to output an author list." H2 says they hold the paper's identity: "which paper this is." H3 says the authors are already there, just not along vocabulary directions, so the readout tools cannot see them.

结论是：三个假设并不互斥，目前各自在不同位置有不同程度的支持。冒号和名字位置存的是任务和格式，H1 成立。H2 部分支持：标题位置能读出论文属性，X1 的跨问法替换也能恢复作者分数，但 M3 不高于 L4 的文字水平，X3 也分不清是论文身份还是语义，而 Qwen 胜过 BERT 按预注册只作参考；另外 X4 最早解码出的是第一作者（L8），年份要到 L40，和 H2 预测的顺序相反。H3 证据不足：连专门用作者提示拟合的读出工具也读不出名字；probe 从 L8 起能读出一点第一作者（+0.046），但只用标题文字的基线不弱于 Qwen（名字首 token 上 BERT +0.088，Qwen +0.018），所以不能排除标题文字的解释。X2 只替换子空间：论文属性子空间几乎搬不动作者信号，不支持 H2；作者子空间按判据通过，但它本身几乎解码不出作者，只搬动完整替换的约 4%，不能作为 H3 的有力证据。

> 补充 9（答辩后新增，分析计划在看结果前写定）把这个 probe 扩展到每位作者的全部名字 token 并按角色分开：L40 姓的第一个 token +0.071 [0.058, 0.085]（目标 + 对照 1,395 篇，剔除标题复用，团队聚

类 95% CI；381_stage13/reports/e4b_alltok.md），高于首字母 +0.044 [0.030, 0.057]（1,590 篇）；名的第一个 token 最强（L40 +0.124、L56 +0.163（1,706 篇））。原 E4b 只打分第一个 token（多半是首字母），所以低估了标题位置携带的名字信息。2026-11-06：X7 补做了另外 4 个假标题，按 5 个平均判定，target、control 和全部论文都满足预注册判据（S(L52) 分别为 +0.022、+0.042、+0.028，均 ≤ 0.05）（补充 10 第 2 项；381_stage13/reports/b1_x7f_report.md）；原来对照组只用 false_0 时为 0.051，属于单个假标题的波动。

Addendum 9 (post-defence, plan fixed before the results) extends the probe to every name token by role: surname first token at L40 +0.071 [0.058, 0.085](1,395 papers; e4b_alltok.md), above the initial +0.044 [0.030, 0.057]; the given-name first token is strongest (L40 +0.124, L56 +0.163). E4b scored only the first token, mostly an initial, so it underestimated the name information at the title.

2026-11-06（批次 1）：X1–X4 改为给全部作者打分（全部名字 token），并按作者排位分层。结论都不变：X1 仍与任务无关，L48 有一点小而确实的任务依赖（作者 − 年份 +0.024 [0.019, 0.030]）；X3 改写标题保留约 90% 的作者偏好，target 与 control 一样；X4 从 L8 起可解码出作者，各作者排位相同，认得出的论文在 L8–L40 更高；但只靠标题文字的基线不弱于 Qwen，仍不能排除标题文字的解释。X2 用全部作者重新拟合作者子空间后，搬动量从 0.016 升到 0.032（完整替换的约 7%），但子空间本身仍几乎解码不出作者（留出 $R^2 \le 0.009$），H3 仍证据不足（补充 10 第 1 项，硬检查 PASS；381_stage13/reports/b1_xall_report.md）。E4b 5 个假标题（偏离 D1 逐个假标题排除，保留 4,654 篇，平均 1.5 个假标题）：L40 姓首 token +0.076 [0.068, 0.083]，姓 − 首字母 +0.033 [0.023, 0.042]，原 E4b 第一个 token +0.066；置换对照首字母仍 +0.018，其余 ≈ 0；硬检查 E4b L40 +0.0609 PASS（补充 10 第 4 项；381_stage13/reports/b1_e4b5_report.md）。按逐个假标题排除、5 个假标题平均，名的第一个 token 从 L16 起、姓从 L24 起 target−control 为正（姓 L40 +0.030 [0.013, 0.048]）；首字母不显著（L40 +0.014 [−0.002, +0.030]）。只用 false_0、样本较小时姓在 L56 前不显著，此前的说法已被取代（b1_e4b5_report.md，per-donor target − control；summary_b1_e4b5/perdonor_by_run_layer.csv Vpd）。P1/P2 五个假标题还在运行。

Batch 1 (2026-11-06): X1–X4 re-scored on all authors and all name tokens, stratified by author position — conclusions unchanged (X1 task-independent, small L48 author − year +0.024 [0.019, 0.030]; X3 keeps about 90%; X4 from L8; X2 with an all-author subspace moves 0.032 but still decodes almost nothing, held-out $R^2 \le 0.009$, so H3 stays insufficient)(b1_xall_report.md). E4b minus five false titles (per-donor exclusion, 4,654 papers): surname first token at L40 +0.076 [0.068, 0.083], surname − initial +0.033; target − control positive from L16 for the given-name first token and from L24 for the surname (L40 +0.030 [0.013, 0.048]); initial not significant (L40 +0.014 [−0.002, +0.030]); with false_0 only the surname is not significant before L56 (earlier statement superseded)(b1_e4b5_report.md). P1 / P2 with five false titles still running.

The conclusion: the three are not mutually exclusive, and so far each has a different degree of support in a different place. The colon and name positions hold the task and format, so H1 holds there. H2 is partly supported: paper attributes are readable at the title position and X1's cross-question patching restores author scores, but M3 is not above the L4 text level, X3 cannot separate paper identity from meaning, and Qwen beating BERT counts only as supporting evidence under the pre-registration; also, X4 decodes the first author first (L8) and the year only at L40, the opposite of H2's predicted order. H3 has insufficient evidence: even a readout fitted on author prompts cannot read the names; a probe

reads a little of the first author from L8 (+0.046), but a title-text-only baseline is not weaker than Qwen (given-name token: BERT +0.088, Qwen +0.018), so title wording cannot be ruled out. X2, patching only subspaces: the paper-attribute subspace moves almost none of the author signal, so H2 is not supported; the author subspace passes its rule but can hardly decode authors itself and moves only about 4% of full patching, so it is not strong evidence for H3.

连起来就是：冒号和名字位置承载任务；标题位置能读出论文属性，但目前还分不清这是“认出了这篇论文”还是“读懂了标题的字面意思”；信息在 L44–52 送进名字位置，到 L57–61 才把名字说出来。X3 已经跑完，结论是否定的：改写标题在 Qwen 中间层（L28–L40，最高 L36 +0.16）比用词相近的近邻 D 更接近原标题，超出 L4 文字水平；行为上，改写保留了原标题对 D 约 90% 的第一作者偏好。但两项检验说明这不足以证明“论文身份”：第一，只懂语义、不认识任何作者的句子向量模型 all-mpnet，在用词真正匹配的子集上也有相当的优势；第二，认不认得出没有任何差别：行为上的 90% 在 target（0.90）和 control（0.91）之间一样，中间层的优势也一样（L36 上 target +0.158、control +0.160，各层两组之差都约为 0）。如果是认出了论文，target 应该更高。要主动说明一点：按补充 4 预注册的字面判据（G′ > 0 且超出 L4，调参集找到、测试集确认），X3 其实判为“存在论文身份信号”；文本模型基线和 target/control 拆分是看到结果后加的检验，我们据此把解读降为“语义表示”。两者都如实报告，这一步让结论更保守，而不是更乐观。所以 X3 只说明 Qwen 中间层表示了标题的意思，而不只是用词，分不清是认出论文还是懂得意思。Qwen 特别的只是时间：优势出现在中间层，后层又消失。总结起来，M3、X3、X4 这三个专门用来区分“认出论文”和“读懂文字或意思”的实验，都没能证明是认出论文。

Putting it together: the colon and name positions carry the task; the title position carries readable paper attributes, but we cannot yet tell whether that is recognising the paper or understanding the title's wording; the information enters the name positions at L44 to 52, and the name is only spoken at L57 to 61. X3 has finished, and the answer is negative: in Qwen's middle layers (L28–L40, peaking at +0.16 at L36) the paraphrased title is closer to the original than a word-similar neighbour D, beyond the L4 text level; behaviourally the paraphrase keeps about 90% of the original title's first-author advantage over D. Two checks show this is not enough to establish paper identity. First, all-mpnet, a sentence-embedding model that understands meaning but knows no authors, shows a comparable advantage on the subset where word overlap is truly matched. Second, recognition makes no difference: the behavioural 90% is the same for target (0.90) and control (0.91), and so is the middle-layer advantage (at L36, target +0.158 and control +0.160; the group difference is about 0 at every layer). If the model were recognising the paper, target should be higher. One point to state up front: under the literal pre-registered criterion in addendum 4 (G′ > 0 and above L4, found on the tune split and confirmed on test), X3 counts as showing a paper-identity signal; the text-model baseline and the target/control split were added after seeing the results, and on that basis we downgrade the reading to "semantic representation". We report both; this makes the conclusion more conservative, not less. So X3 shows that Qwen's middle layers represent the title's meaning, not just its words, but cannot separate recognising the paper from understanding the meaning. What is distinctive about Qwen is only the timing: the advantage appears in the middle layers and fades later. Taken together, M3, X3 and X4, the three experiments designed to separate recognition from wording or meaning, all fail to show recognition.